

Nisha Saxena

Chief Substation Engineer
UPPCL · 14 years

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PROFESSIONAL SUMMARY

Dedicated power sector professional with 14 years of hands-on experience specializing in Protection Relay Coordination and Circuit Breaker Testing. Passionate about India's energy transition and digital transformation of the power sector.

CORE COMPETENCIES

• Harmonic Analysis	• Protection Relay Coordination	• Switchgear Selection
• Circuit Breaker Testing	• Single Line Diagram	• RTU Programming
• PLC Programming (Siemens S7/Allen Bradley)	• Earthing Design	• HVDC Systems
• Grid Code Compliance	• Demand Response Management	• Energy Trading
• Power Market Operations	• Machine Learning	• AMI Systems

PROFESSIONAL EXPERIENCE

Chief Substation Engineer UPPCL	2022 – Present
■ 132kV Transmission Line (20 km) — Tower design, foundation engineering, stringing and commissioning for interstate grid connectivity ■ 100 MW Wind Energy Project — Turbine foundation design, inter-array cabling, substation and SCADA implementation ■ 132kV Transmission Line (11 km) — Tower design, foundation engineering, stringing and commissioning for interstate grid connectivity ■ 100 MW Wind Energy Project — Turbine foundation design, inter-array cabling, substation and SCADA implementation ■ Improved transformer life expectancy by 5 years through online DGA monitoring	
Power Systems Engineer Torrent Power	2017 – 2022
■ Substation Automation — IEC 61850 based protection and control implementation at 19 EHV substations ■ Power Quality Improvement — Harmonic filter design and installation at Ahmedabad reducing THD from 12% to 3.5% ■ Demand Response Program — Implementation across 12 industrial consumers achieving peak demand reduction of 7 MW ■ Filed 14 patents related to power system optimization	
Grid Stability Analyst KEC International	2016 – 2017
■ BESS Installation (14 MWh) — Grid-scale battery storage commissioning with frequency regulation and peak shaving applications ■ Protection Relay Replacement Program — Digital relay retrofitting across 13 substations improving fault clearance time by 40% ■ Load Forecasting Model — ML-based short-term load forecasting achieving 4% MAPE for a 2 MW utility ■ Distribution Network Strengthening — R-APDRP project covering 3 feeders with automated fault detection and restoration ■ Delivered EPC project 17% under budget saving ■11 Cr	
Grid Modernization Engineer BHEL	2011 – 2016
■ OT Cybersecurity Hardening — ICS security assessment and remediation across 5 substations per IEC 62351 standards	

- Load Forecasting Model — ML-based short-term load forecasting achieving 17% MAPE for a 10 MW utility
- 132kV Transmission Line (9 km) — Tower design, foundation engineering, stringing and commissioning for interstate grid connectivity

EDUCATION

M.Tech (Energy Technology)

2009

Delhi Technological University

CERTIFICATIONS & TRAINING

- ✓ LEED Green Associate
- ✓ AutoCAD Certified Professional
- ✓ GE Grid Solutions Certified

References available upon request · Last updated: February 2026